Tennis genetic algorithm

Group - Iga Świątek Fanklub

Magdalena Rąpała
Miłosz Liniewiecki

AEI faculty

24.06.2026

Agenda

Introduction to problem and environment

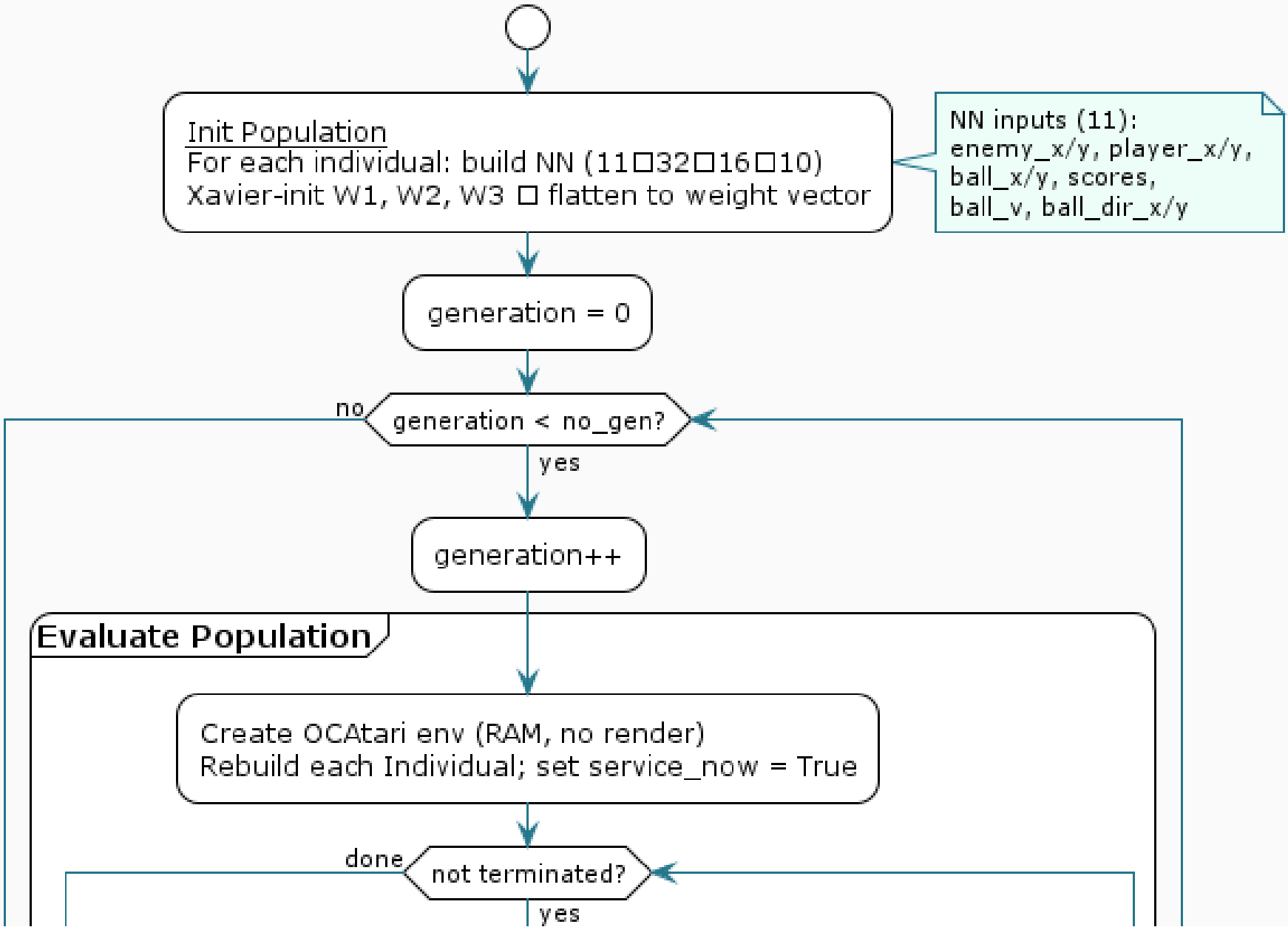
Tennis is an Atari Gymnassium environment with two players.

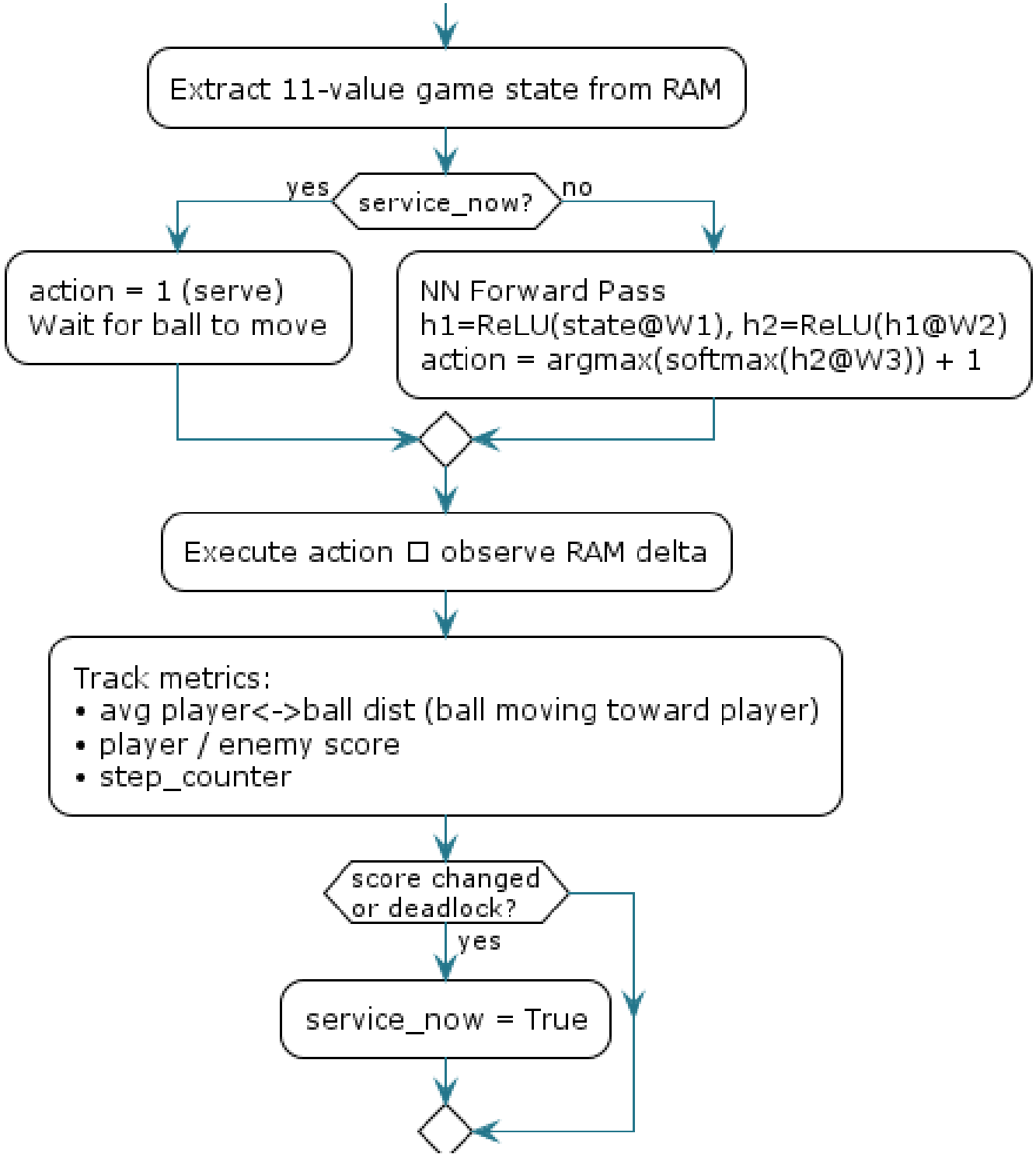
- Blue player - Game AI, our enemy.
- Orange player - Our AI, the one we trained.

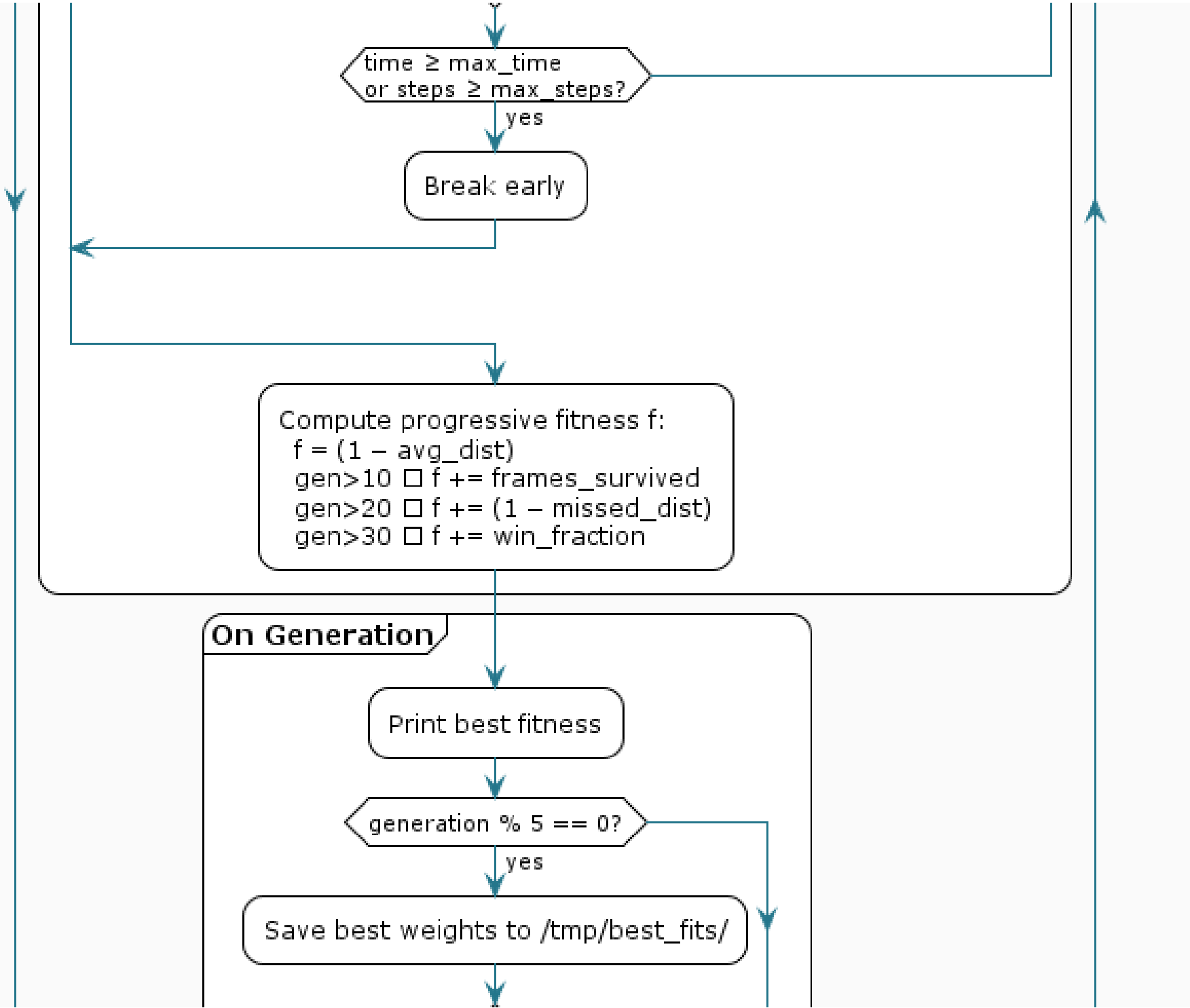
Atari Gymnassium does not give you any data to feed into the genetic algorithm, which is why we used OCAtari for RAM values extraction.

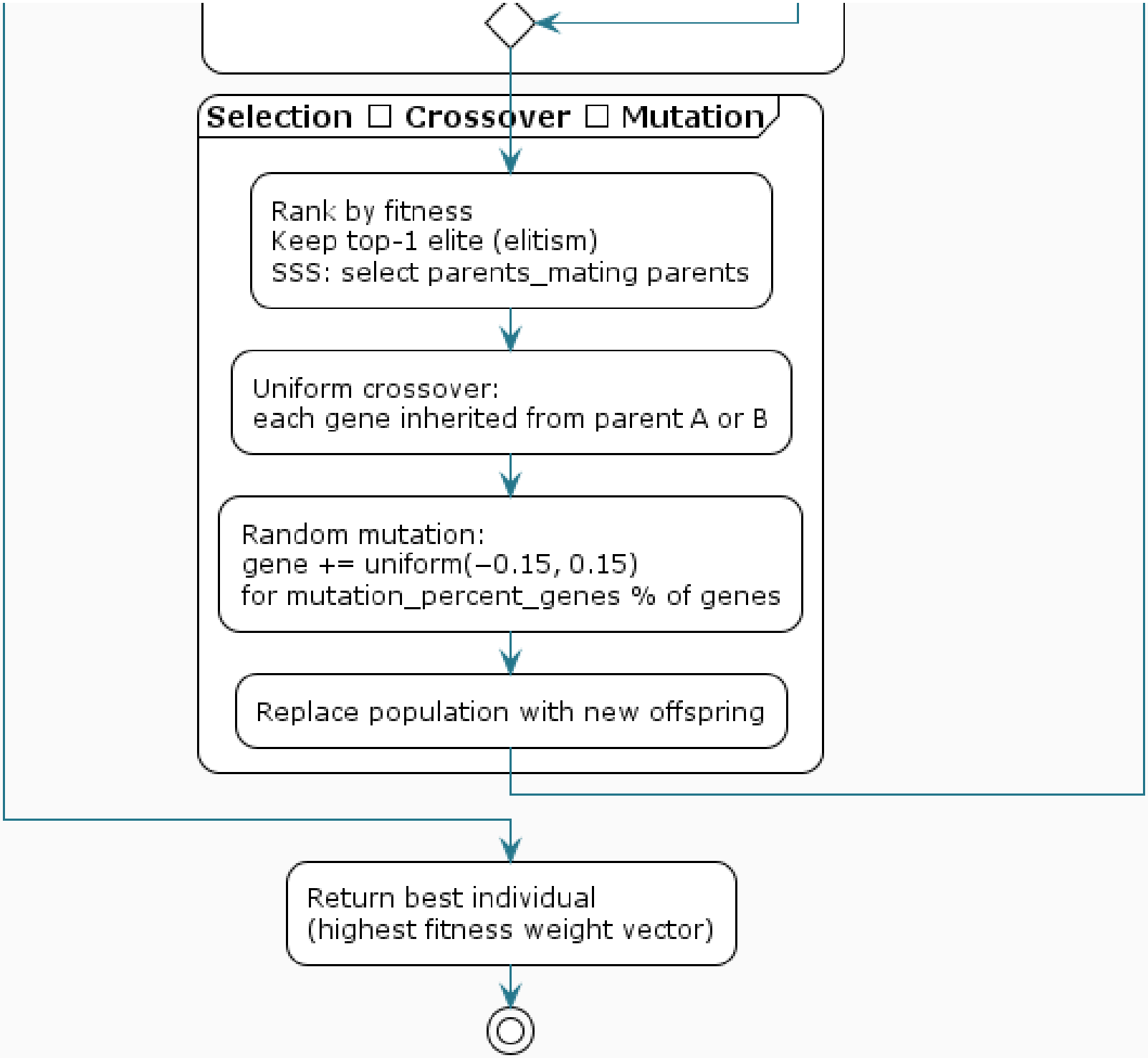
Figure: Tennis environment during training.

Tennis Genetic Algorithm Flowchart




Extract 11-value game state from RAM
service_now?
yes
no
action = 1 (serve)
Wait for ball to move
NN Forward Pass
h1=ReLU(state@W1), h2=ReLU(h1@W2)
action = argmax(softmax(h2@W3)) + 1
Execute action □ observe RAM delta
Track metrics:
• avg player<->ball dist (ball moving toward player)
• player / enemy score
• step_counter
score changed
or deadlock?
yes
service_now = True


time ≥ max_time
or steps ≥ max_steps?
yes
Break early
Compute progressive fitness f:
f = (1 − avg_dist)
gen>10 ☐ f += frames_survived
gen>20 ☐ f += (1 − missed_dist)
gen>30 ☐ f += win_fraction
On Generation
Print best fitness
generation % 5 == 0?
yes
Save best weights to /tmp/best_fits/


Selection ☐ Crossover ☐ Mutation
Rank by fitness
Keep top-1 elite (elitism)
SSS: select parents_mating parents
Uniform crossover:
each gene inherited from parent A or B
Random mutation:
gene += uniform(−0.15, 0.15)
for mutation_percent_genes % of genes
Replace population with new offspring
Return best individual
(highest fitness weight vector)

Pseudocode 1/6 - Init

```
FUNCTION init_population(no_inv):
population <- []

FOR i = 1 TO no_inv DO
        Build NN with layers: n_x=11 -> h1=32 -> h2=16 -> n_y=10
        Xavier-init weight matrices W1, W2, W3
        population.append(flatten([W1, W2, W3]))
END FOR

RETURN population
```

Pseudocode 2/6 - Individual init

```
population <- init_population(no_inv)
generation <- 0

WHILE generation < no_gen DO

generation <- generation + 1

FOR each individual in population DO

        Create OCAtari "Tennis-v4" env (RAM mode, no render)
        Rebuild Individual from weight vector
        service_now <- True
```

Pseudocode 3/6 - State extraction

```
WHILE not terminated DO
        state <- extract 11-value game state from RAM
        IF service_now THEN
                action <- 1  (force serve)
                IF ball started moving THEN
                        service_now <- False
                END IF

        ELSE
                h1     <- ReLU(state @ W1)
                h2     <- ReLU(h1    @ W2)
                logits <- h2 @ W3
                action <- argmax(softmax(logits)) + 1
        END IF
        Execute action in env -> observe RAM delta
```

Pseudocode 4/6 - Safety checks

```
Track metrics:
        avg_player_ball_dist  <- distance(player, ball)
                when ball moves toward player
        player_score, enemy_score
        step_counter <- step_counter + 1

IF score changed OR deadlock detected THEN
        service_now <- True
END IF

IF elapsed_time >= max_time
        OR step_counter >= max_steps THEN
        BREAK  (early termination)
END IF

END WHILE
```

Pseudocode 5/6 - Fitness calculation

```
        Compute fitness f:
        f <- 1 - avg_player_ball_dist
        IF generation > 10 THEN  f <- f + (step_counter / max_steps)
        IF generation > 20 THEN  f <- f + (1 - avg_missed_distance)
        IF generation > 30 THEN  f <- f + (player_score / total_points)

        Store f as fitness of this individual

END FOR
```

Pseudocode 6/6 - Selection, best result

```
Print best fitness of current generation

IF generation MOD 5 = 0 THEN
        Save best weight vector to /tmp/best_fits/gen-{g}-fit-{f}.npy
END IF

Rank population by fitness
elite <- top-1 individual  (preserved unchanged)

parents_mating <- floor(no_inv x parents_frac)
parents <- SSS_select(population, parents_mating)

new_population <- [elite]
```

Results

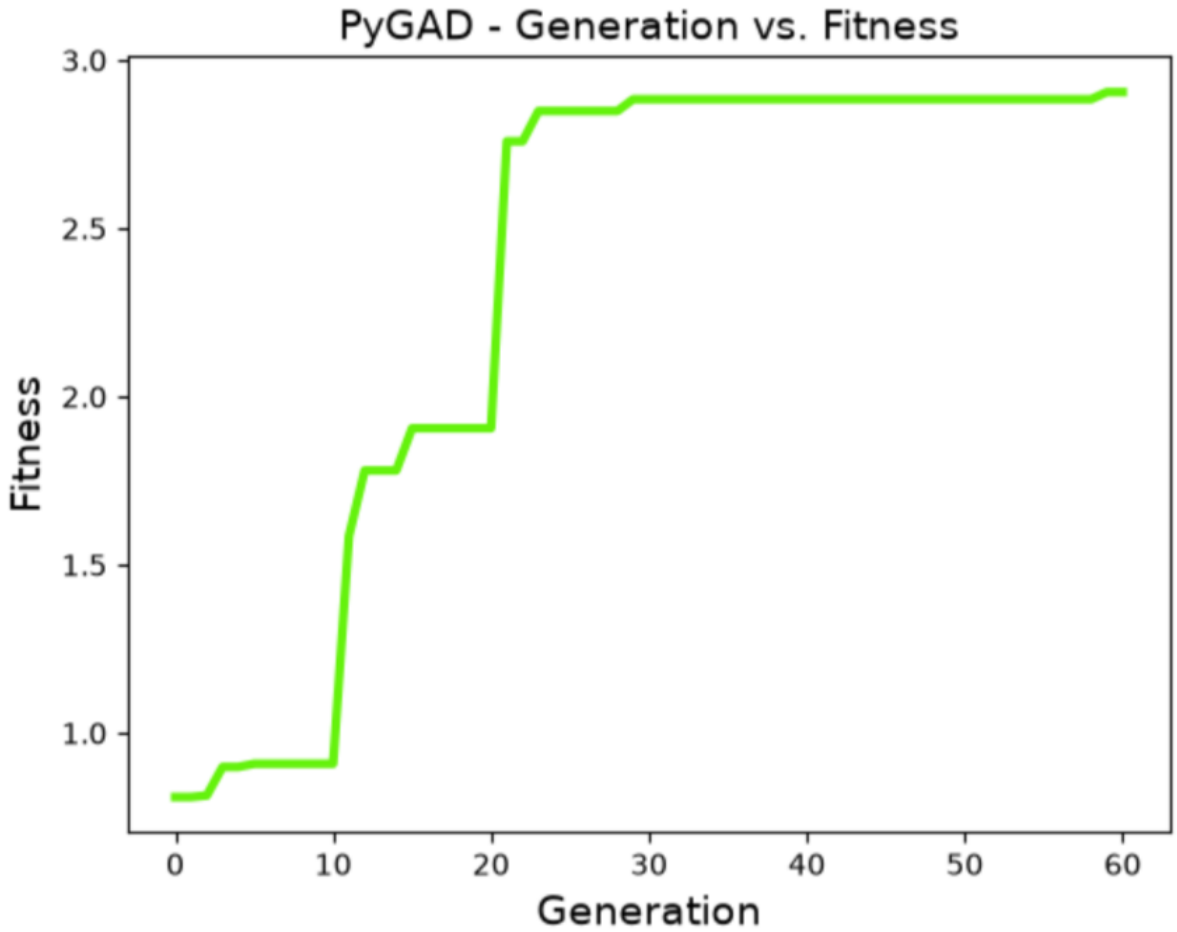


Figure: Results